

Integration

Week 6A

Overview

Integration

Key Ideas - *Integration by Simple Substitution Further Functions*

Key Idea - Simple Substitution Further Functions

As we have seen there are occasions when it is possible to perform an apparently difficult integration by first making a substitution to simplify the integrand. This has the effect of changing the variable and the integrand. In this lesson we will meet several other examples of this type, such as

$$\int e^x \sqrt{e^x + 1} \, dx \quad , \quad \int \cos x e^{\sin 2x} \, dx \quad , \quad \int \frac{4x^3 + 2x}{x^4 + x^2} \, dx$$

Be aware that sometimes an apparently sensible substitution does not lead to an integral that you will be able to evaluate. You must then need to try out alternative substitutions.

Given an integral of the form $\int f(g(x))g'(x) dx$

Step 1: Let $u = g(x)$

Step 2: Find $\frac{du}{dx} = g'(x)$, so $du = g'(x) dx$.

[You may have to adjust constant factors]

Step 3: Make the substitutions $u = g(x)$ and $du = g'(x) dx$ in the original integral, to get $\int f(u) du$.

The integral must now be entirely in terms of u , including the ' dx '.

Step 4: Evaluate the new integral (with respect to u) $\int f(u) du$.

Step 5: In the answer back substitute each occurrence of variable u by $g(x)$.

The answer must now be entirely in terms of x , or the original variable.

Example: Integrate $\int e^x \sqrt{e^x + 1} \, dx$.

In this example we make the substitution $u = e^x + 1$, in order to simplify the square root term.

Step 1: Let $u = e^x + 1$

We shall see that the rest of the integrand, $e^x \, dx$, will be taken care of automatically in the substitution process, and that this is because e^x is the derivative of that part of the integrand used in the substitution, i.e. $u = e^x + 1$

Step 2: $\frac{du}{dx} = e^x$ and $du = e^x \, dx$

So, substitute u for $e^x + 1$, and with $e^x \, dx = du$ in $\int e^x \sqrt{e^x + 1} \, dx$

$$\begin{aligned} \text{Step 3: } \int e^x \sqrt{e^x + 1} \, dx &= \int \sqrt{e^x + 1} e^x \, dx \\ &= \int \sqrt{u} \, du \end{aligned}$$

Now integrate the new function with respect to u .

$$\begin{aligned}
 \text{Step 4: } \int \sqrt{u} \, du &= \int u^{1/2} \, du \\
 &= \frac{u^{3/2}}{3/2} + c \\
 &= \frac{2u^{3/2}}{3} + c
 \end{aligned}$$

We now back substitute each occurrence of variable u by $e^x + 1$, giving

$$\text{Step 5: } \frac{2(e^x + 1)^{3/2}}{3} + c \text{ or } \frac{2}{3} \sqrt{(e^x + 1)^3} + c$$

Note: Check this solution by differentiation $\frac{d}{dx} \left(\frac{2}{3} (e^x + 1)^{3/2} \right)$

Example: Integrate $\int 3 \cos 2x e^{\sin 2x} dx$

In this example we make the substitution $u = \sin 2x$, in order to simplify the square root term.

Step 1: Let $u = \sin 2x$

We shall see that the rest of the integrand, $3 \cos 2x dx$, will be taken care of automatically in the substitution process, and that this is because $\sin 2x$ is the derivative of that part of the integrand used in the substitution, i.e. $u = \sin 2x$

$$\text{Step 2: } \frac{du}{dx} = 2 \cos 2x \text{ and } du = 2 \cos 2x dx \Rightarrow \frac{du}{2} = \cos 2x dx$$

So, substitute u for $\sin 2x$, and with $\cos 2x dx = \frac{du}{2}$ in $\int 3 \cos 2x e^{\sin 2x} dx$

$$\begin{aligned} \text{Step 3: } \int 3 \cos 2x e^{\sin 2x} dx &= 3 \int \cos 2x e^{\sin 2x} dx \\ &= 3 \int e^u \frac{du}{2} \\ &= \frac{3}{2} \int e^u du \end{aligned}$$

Now integrate the new function with respect to u .

$$\begin{aligned}\text{Step 4: } \frac{3}{2} \int e^u du &= \frac{3}{2} e^u + c \\ &= \frac{3}{2} e^{\sin 2x} + c\end{aligned}$$

We now back substitute each occurrence of variable u by $\sin 2x$, giving

$$\text{Step 5: } \frac{3}{2} e^{\sin 2x} + c$$

Note: Check this solution by differentiation $\frac{d}{dx} \left(\frac{3}{2} e^{\sin 2x} + c \right)$

Example: Integrate $\int \frac{4x^3 + 2x}{x^4 + x^2} dx$

This is a special case of a commonly occurring form,

$$\frac{\text{derivative of denominator}}{\text{denominator}}$$

where the integral is the **logarithm** of the denominator.

In this example we make the substitution $u = x^4 + x^2$, in order to simplify.

Step 1: Let $u = x^4 + x^2$

We shall see that the rest of the integrand, $4x^3 + 2x \, dx$, will be taken care of automatically in the substitution process, and that this is because $4x^3 + 2x$ is the derivative of that part of the integrand used in the substitution, i.e. $u = x^4 + x^2$.

Step 2: $\frac{du}{dx} = 4x^3 + 2x$ and $du = 4x^3 + 2x \, dx$

So, substitute u for $x^4 + x^2$, and with $du = 4x^3 + 2x \, dx$ in $\int \frac{4x^3 + 2x}{x^4 + x^2} dx$

$$\text{Step 3: } \int \frac{4x^3 + 2x}{x^4 + x^2} dx = \int \frac{1}{u} du$$

Now integrate the new function with respect to u .

$$\text{Step 4: } \int \frac{1}{u} du = \ln|u| + c$$

We now back substitute each occurrence of variable u by $x^4 + x^2$, giving

$$\text{Step 5: } = \ln|x^4 + x^2| + c$$

Note: Check this solution by differentiation $\frac{d}{dx} (\ln|x^4 + x^2|)$

Example: Integrate $\int \cos x \sin^6 x \, dx$.

In this example we first rewrite $\int \cos x \sin^6 x \, dx = \int \cos x (\sin x)^6 \, dx$.

Then we make the substitution $u = \sin x$, in order to simplify the square root term.

Step 1: Let $u = \sin x$

We shall see that the rest of the integrand, $\cos x \, dx$, will be taken care of automatically in the substitution process, and that this is because $\cos x$ is the derivative of that part of the integrand used in the substitution, i.e. $u = \sin x$

Step 2: $\frac{du}{dx} = \cos x$ and $du = \cos x \, dx$

So, substitute u for $\sin x$, and with $\cos x \, dx = du$ in $\int \cos x \sin^6 x \, dx$

$$\begin{aligned} \text{Step 3: } \int \cos x \sin^6 x \, dx &= \int \cos x (\sin x)^6 \, dx \\ &= \int u^6 \, du \end{aligned}$$

Now integrate the new function with respect to u .

$$\text{Step 4: } \int u^6 du = \frac{u^7}{7} + c$$

We now back substitute each occurrence of variable u by $\sin x$, giving

$$\text{Step 5: } \frac{(\sin x)^7}{7} + c$$

Note: Check this solution by differentiation $\frac{d}{dx} \left(\frac{(\sin x)^7}{7} + c \right)$

Exercise: Find the following:

(a) $\int \sin x \sqrt{\cos x} \, dx$

$$(b) \int x \cos x^2 e^{\sin x^2} dx$$

(c) $\int \frac{4e^x + e^x}{2e^{2x} + e^x + 1} dx$

(d) $\int \tan 2x \sec^2 2x \, dx$